TECHNICAL REPORT / ITU UAE HACKATHON 2026

MedRoute X

Evidence-grounded multi-agent coordination for synthetic medicine continuity

Release: v0.10.0 (09 November 2026). **Code:** github.com/cazantzevaol/MedRoute-X. **System:** offline deterministic ten-role prototype; optional live LLM adapter has NOT been evaluated. Final submission remains with the entrant.

1. Problem statement and intended use

Shortages can create a coordination problem between healthcare facilities, stock holders, suppliers, logistics and regulators. Actors may receive conflicting or delayed information. MedRoute X studies whether bounded role-specific information, progressive disclosure, independent operational validation and source-linked policy context can improve **synthetic** decision processes.

The prototype does **not** manage real inventory, process patient records, select treatment, authorize controlled-medicine movement, or implement UAE regulator decisions. Candidate transfers are simulations and always remain pending human authority.

2. Evaluation scenario: sequential incomplete information

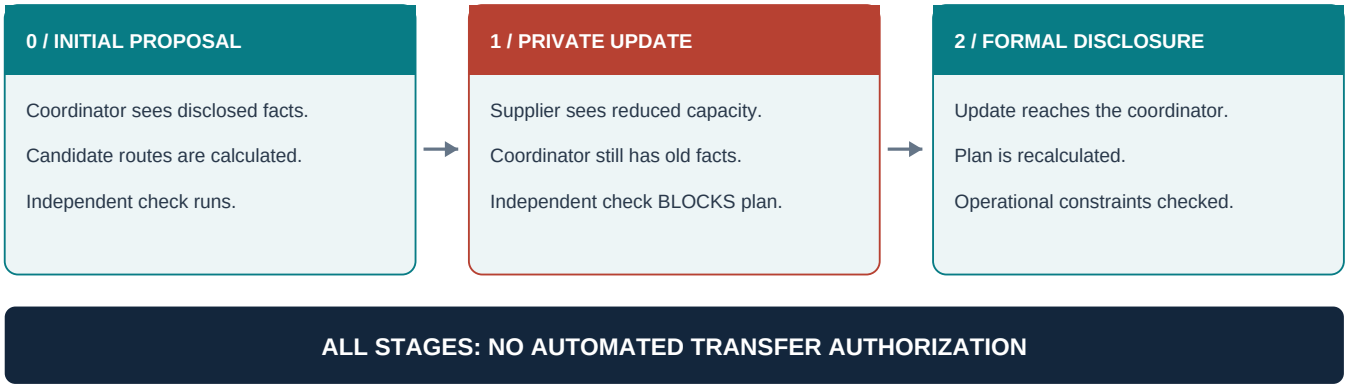

Figure 1. The built-in three-state disclosure scenario. A separate controlled-medicine emergency case indicates an EDE-specific review pathway without granting a permit.

All identities, quantities, routes and urgency weights are invented. The workflow measures protocol correctness and optimization metrics, not patient outcomes.

ARCHITECTURE AND STANDARDS

3. Multi-agent coordination and boundaries

Ten role-specific offline assessors: Coordinator, Hospital, Inventory, Demand, Supplier, Logistics, Regulator, GlobalPolicy, Equity and Auditor. Each receives a bounded view; handoffs are typed and auditable. The independent operational validator checks stock reserve, route capacity, temperature and timing. Policy retrieval supplies research context, not permission. Role findings quantify visible shortages, reserve-protected surplus, demand and route constraints; private supplier details are withheld until formal disclosure.

Human authorization: neither an offline assessor nor an optional LLM response can issue a regulatory permit. Human authorization is always required. The browser demo runs on loopback; it has no user authentication and is not a deployed medical service.

4. Mapping to ITU-T Y.3172 (architectural analogy)

Y.3172-oriented element	MedRoute X analogue	Limit
Data collection / preparation	Synthetic inventory, route and event fixtures; scoped observations.	No live healthcare data pipeline.
Model and pipeline functions	Optimization, retrieval, role evaluation, deterministic policy gate.	Not an ITU-certified implementation.
Orchestration and sandbox	Sequenced events, typed handoffs, replayable tests.	Offline prototype; no network orchestration.
Monitoring and outputs	Audit hash chain, evidence references, regression checks.	Hash chain is not tamper-proof against a full rewrite.

5. Mapping to ITU AI Readiness Framework 2.0

Dimension	Implemented signal / identified need
Data governance and sovereignty	No patient data; public source metadata; future expert-reviewed national corpus needed.
Human oversight and accountability	Human approval gate; role-specific audit; no delegated legal decisions.
Digital infrastructure and skills	Local Python demo, reproducible tests; trained operators and secure hosting not yet available.
Inclusion and contextualization	Synthetic priority-weighted allocation; fairness and local applicability unvalidated.

The standards mappings are qualitative design interpretations, not conformity assessments.

POLICY EVIDENCE AND HUMAN AUTHORITY

6. Local policy mapping (UAE)

The curated knowledge base has 14 source pointers and 18 short claims. These are manually summarized metadata and selected official excerpts, **not** a complete national legislative corpus. Every candidate medicine movement is marked HUMAN_REVIEW_REQUIRED.

Official source / reference	Relevant scope	Prototype response
EDE emergency transfer service [R1]	Request for approval of emergency transfers of narcotic, psychotropic and semi-controlled medicines between eligible facilities.	Explicit controlled + emergency scenario -> EDE_EMERGENCY_TRANSFER_REVIEW_REQUIRED; no permit.
UAE Federal Law 2/2019, Arts 4 and 16 [R2]	Confidentiality and conditions/exceptions concerning health information.	Only synthetic data; no patient-specific document ingestion.
UAE Federal Law 2/2019, Art 13 [R2]	Conditions restricting certain offshore health-data processing/transfers.	No claim that this sandbox is approved for real health data.
UAE Federal Decree-Law 38/2024 [R3]	Medical products and pharmaceutical establishments (official legislation metadata).	Product/facility authorization requires qualified review.
EDE supply-chain announcement, 24 Feb 2026 [R7]	Multiple authorized agents per medical product; supply continuity policy context.	Research context only; not a facility-to-facility transfer permit.
EMA shortage solidarity example [R4]	Cross-border EU/EEA shortage coordination (comparative context).	Research comparator only; no UAE legal force.

Critical distinction: the EDE service lists patient-related supporting documents for certain applications. Those records belong only in the competent authorized channel, never in MedRoute X. Screening of public document ingestion is heuristic and does not guarantee de-identification.

7. Evidence status and interpretation

English excerpts of Federal Law 2/2019 Articles 4, 13 and 16 were checked against the official UAE legislation portal. The authoritative Arabic wording, implementing rules, scope, current amendments and institution-specific procedures require independent expert verification. A missing source is an **evidence gap**, not a proven defect in UAE law.

EVALUATION AND REPRODUCIBILITY

8. Results on 30 seeded synthetic cases

Reproducible offline benchmark (seed 2026). Lower values are better. The urgency weights are synthetic institutional preferences and must not be interpreted as medical triage or patient harm reduction.

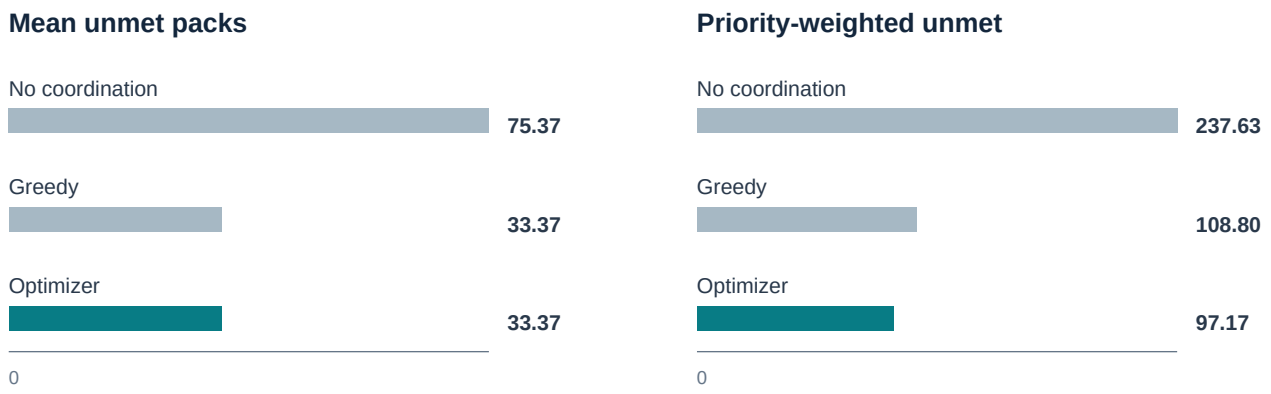

Figure 2. Reproduced from reports/benchmark_summary.json. All three methods passed operational validation in all 30 cases; this is not legal clearance.

Measured outcome: the optimizer reduces synthetic priority-weighted unmet demand by 10.69% versus greedy. **Total unmet packs are unchanged** versus greedy. No patient outcomes or real regulatory compliance were measured.

9. Verification and auditability

Offline regression suite: 376 tests in the 09 November 2026 local release suite. Protocol acceptance: 30 synthetic scenarios, 300 role assessments, valid hash-chain checks and zero autonomous legal approvals in the previous v0.9 acceptance run. A hash chain detects inconsistent local records; it is not an external trust anchor.

New release gate inspects PDF page count and version, mandatory ITU topics, local knowledge-base integrity, benchmark and demo artifacts. It explicitly refuses to infer team registration, GitHub accessibility or formal submission from local files.

10. Decision support and limitations

A review packet compares keep-stock, greedy and priority-aware proposals, excludes operationally invalid options and identifies remaining supply requests and human review tasks. Local feedback only breaks ties between equally effective valid options in the same scenario context; it does not retrain a model or override safety gates.

The ten offline roles are deterministic rule-based prototypes, **not ten independent LLMs**. Optional model-backed code is unvalidated and never used in these results. No live supplier integration or trained adaptive model is claimed. There is no independent legal annotation gold set, multilingual corpus, clinical validation, real data, secure user authentication, or approved production deployment.

POLICY GAPS, REFERENCES AND NEXT STEPS

11. Policy and standards gaps (research hypotheses)

G1: Institutional escalation. Under incomplete information, which verified legal and operational actors should receive shortage alerts, and in what order? Need: authoritative role and process documentation plus pharmacy/regulator review.

G2: Machine-readable permissions. How can a policy knowledge base encode jurisdiction, source version, exceptions, responsible authority and review status without mistaking retrieval for permission? Need: legal expert gold set and source provenance verification.

G3: Cross-border comparators. Which EMA/WHO coordination practices could inform future policy design while remaining clearly nonbinding in the UAE? Need: domain-specific gap analysis and international comparison, not automatic importation of foreign rules.

12. Submission readiness (as of 09 November 2026)

Item	Current status
Technical report	Five-page v0.10 draft with Y.3172/readiness/policy/evaluation mappings.
Knowledge base	14 source pointers, 18 curated claims; not a complete or legally validated corpus.
GitHub repository	Public: github.com/cazantzevaol/MedRoute-X. GitHub automated checks passed.
Team registration	Participant reports completed registration; final entrant details remain to be confirmed.
Submission	NOT submitted; official deadline 10 November 2026.
Demo video	Optional at first round (up to 7 minutes), required if shortlisted; no recording claimed.

13. Reproduction and references

Commands: `python -m pip install -r requirements.txt`; `python -m pytest -q`; `python scripts/submission_gate.py --version 0.10`; `python -m medroute.server`. No API key needed for offline runs.

[R1] Emirates Drug Establishment, emergency transfer service: ede.gov.ae / emergency drugs service.
[R2] UAE legislation portal, Federal Law 2/2019: official PDF (Arts 4, 13, 16).
[R3] UAE legislation portal, Federal Decree-Law 38/2024: official metadata.
[R4] European Medicines Agency: shortages steering group.
[R5] ITU: hackathon requirements, FAQ and deadline.
[R6] ITU-T: Y.3172.
[R7] EDE (24 Feb 2026): supply-chain announcement.

Team name and member identities must be confirmed by the entrant. This document does not represent organizer acceptance or regulator endorsement.